

Gram-first predictable-baseline support collapse and low-Hermite Wick boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-095 showed that replacing the rootwise future-feedback reserve by a fraction of the global terminal square creates a sign-indefinite moving-prefix defect. The honest successor must therefore preserve the complete R-077 and R-079 packets before applying the R-085/R-086 rational Gram decomposition. This note performs that reordering and identifies the smallest remaining analytic target.

There are five exact advances. First, the R-079 rational row embeds into the R-085/R-086 translated-Wick decomposition once, with the endpoint square, Wick trace, and R-063 forest assigned to unique owners. Second, spatial-first and root-first Doob decompositions agree only after an exact transfer defect is retained. Third, after the complete R-077 fresh-root cancellation, the genuine coefficient-dominant $T_{Q,k}^>$ and $T_{G,k}^>$ regions of every predictable baseline are empty by Fourier support; a fixed boundary collar is payable. Fourth, a raw quadratic Wick contraction sees only coordinate Hermite ranks zero, one, and two of an arbitrary same-root coefficient. Fifth, the conditional coefficient means occupy different low-Hermite sectors: the raw centered Wick term sees rank two, while q and the transformed mean r see rank one.

These reductions do not prove payability at the moving adapted base. Hermite orthogonality gives no spatial or root-scale gain, and Gaussian integration by parts differentiates later adapted controls. The remaining target is one global expectation-inside lower form bound for all predictable baselines, with the non-rational heat, low, trace, and forest owners retained once. Complete H_N , REG, progressive/revisit H_A , OVERLAP_{src}, Nelson, removals, a measure, and Sector A remain open. Tier stays T4.

The binding authorities are R-063 for the complete coefficient-jet forest, R-077 for complete-packet fresh-root cancellation, R-079 for the full safe current packet, R-085 for the five unshifted and shifted-Hessian families, R-086 for the translated-Wick normal form and payload split, and R-095 for the moving-prefix and mean-ownership boundary.

2 Complete cancellation must precede rational projection

Put

$$Z_0 = U + A^0, \quad a = A^* - A^0, \quad G = DZ_0, \quad c = Da, \quad (2.1)$$

and

$$B_0 = \overline{B}(Z_0), \quad B_1 = \overline{B}(Z_0 + a), \quad Q = G^{\otimes 2} - \Gamma_J. \quad (2.2)$$

The third Taylor remainder and its second-order truncation are

$$\begin{aligned} L &= B_1 - B_0 - D\overline{B}(Z_0)[a] - \frac{1}{2}D^2\overline{B}(Z_0)[a, a], \\ B_T &= B_0 + D\overline{B}(Z_0)[a] + \frac{1}{2}D^2\overline{B}(Z_0)[a, a]. \end{aligned} \quad (2.3)$$

Write the five unshifted R-085 families as

$$\begin{aligned}\mathcal{U}_R &= \frac{1}{2}D\bar{B}(Z_0)[a] : \mathcal{Q} + \frac{1}{4}D^2\bar{B}(Z_0)[a, a] : \mathcal{Q} \\ &\quad + G^T B_0 c + G^T D\bar{B}(Z_0)[a]c + \frac{1}{2}G^T D^2\bar{B}(Z_0)[a, a]c.\end{aligned}\tag{2.4}$$

Theorem 2.1 (once-only rational-row embedding)

At finite cutoff and positive floor, pathwise,

$$\Delta\mathcal{W}_R = \mathcal{U}_R + \mathcal{H}_R + \frac{1}{2}c^T B_1 c,\tag{2.5}$$

and the exact R-086 normal form gives

$$\begin{aligned}\mathcal{H}_R + \frac{1}{2}c^T B_1 c &= \frac{1}{2}L : \mathcal{Q},^a + \frac{1}{2}c^T B_T c \\ &= \mathcal{P}_R^{\text{pay}} + T_Q^> + T_G^> + \frac{1}{2}c^T B_1 c.\end{aligned}\tag{2.6}$$

Consequently the exact nonlinear rational-row extraction Π_R satisfies

$$\Pi_R\left(\mathfrak{L}_\ell + \sum_j(\mathfrak{F}_j + \mathfrak{J}_j)\right) = \mathbb{E}\mathcal{U}_R + \mathbb{E}\mathcal{P}_R^{\text{pay}} + \mathbb{E}T_Q^> + \mathbb{E}T_G^> + \frac{1}{2}\mathbb{E}c^T B_1 c.\tag{2.7}$$

Thus the full safe packet is

$$\begin{aligned}\mathbb{E}\mathfrak{B}_{J,\text{safe}}^> &= (I - \Pi_R)\left[\mathfrak{L}_\ell + \sum_j(\mathfrak{F}_j + \mathfrak{J}_j)\right] \\ &\quad + \mathbb{E}\mathcal{U}_R + \mathbb{E}\mathcal{P}_R^{\text{pay}} + \mathbb{E}T_Q^> + \mathbb{E}T_G^> + \frac{1}{2}\mathbb{E}c^T B_1 c - \mathbb{E}\Delta\mathcal{R}_\leq.\end{aligned}\tag{2.8}$$

Proof. Equation (2.5) is the exact R-085 seven-family Taylor partition. R-086 (3.7) groups the shifted-Hessian pair and endpoint square into the first line of (2.6), and R-086 (6.3) partitions its spatial frequencies into the payable and coefficient-dominant branches. Applying the rational-row extraction to the exact R-079 identity gives (2.7), and adding the untouched complement gives (2.8). No inequality has been used. The Wick trace is already inside $\mathcal{Q}$, while the R-063 forest reconstructs the heat-lifted coefficient in (2.5). Appending either again would duplicate an owner. $\square$

This identity also fixes the legal order inherited from R-077. If

$$\Phi_k = \mathfrak{P}_k + \sum_{r=k}^J \mathfrak{G}_{r,k},\tag{2.9}$$

the complete Wick/trace/forest theorem gives

$$\mathbb{E}\sum_k \sum_{r=k}^J \mathfrak{G}_{r,k} = 0.\tag{2.10}$$

The cancellation is an assertion about the complete endpoint packet, not about its rational row or an individual Hermite chaos. Hence (2.10) must be used before applying Π_R to each predictable baseline $\mathfrak{P}_k$.

3 Spatial-first Doob ledger and the product defect

Let

$$\rho = Q_{\text{II}}^{1/2} M(Z_0 + a)^T c, \quad \|\rho\|^2 = c^T B_1 c. \quad (3.1)$$

For $s = \ell$ put $D_s = P_\ell$, and for a probability root put $D_s = d_j$. Define

$$\mathcal{D}_s(X, Y) = \mathbb{E}\langle D_s X, D_s Y \rangle. \quad (3.2)$$

For the R-086 factors write

$$C_{t,r,m}^Q = (\Delta_m K_t)[P_r], \quad Z_n^Q = \Delta_n \mathcal{Q}, \quad C_{t,r,m}^G = (\Delta_m K_t)[H_r], \quad Z_n^G = \Delta_n G. \quad (3.3)$$

With separate resonance and payload widths, set

$$\begin{aligned} \mathcal{K}_s^> &= \frac{1}{2} \int_0^1 (1-t) \sum_{\substack{r, |m-n| \leq L_{\text{res}} \\ m > r + L_{\text{gap}}}} \mathcal{D}_s(C_{t,r,m}^Q, Z_n^Q) dt \\ &+ \int_0^1 (1-t) \sum_{\substack{r, |m-n| \leq L_{\text{res}} \\ m > r + L_{\text{gap}}}} \mathcal{D}_s(Z_n^G, C_{t,r,m}^G) dt + \frac{1}{2} \mathbb{E} \|D_s \rho\|^2. \end{aligned} \quad (3.4)$$

Cross-Doob Pythagoras gives the exact terminal ledger

$$\boxed{\mathbb{E} \mathcal{P}_R^> = \mathcal{K}_\ell^> + \sum_{j=j_0}^J \mathcal{K}_j^>}. \quad (3.5)$$

In particular $d_j \rho = \zeta_j + y_j$, so

$$\frac{1}{2} \mathbb{E} \|d_j \rho\|^2 = \frac{1}{2} \mathbb{E} \|y_j\|^2 + \frac{1}{2} \mathbb{E} \|\zeta_j\|^2 + \mathbb{E} \langle \zeta_j, y_j \rangle. \quad (3.6)$$

Thus the spatial-first ledger preserves, rather than removes, the R-095 moving-prefix defect.

Proposition 3.1 (Doob/product commutator)

Deterministic Littlewood–Paley projections commute with probability conditioning, $[d_j, \Delta_m] = 0$. A bilinear product does not. For a continuous bilinear map $\mathfrak{m}$ put

$$\text{Cov}_j^{\mathfrak{m}}(X, Y) = P_j \mathfrak{m}(X, Y) - \mathfrak{m}(P_j X, P_j Y). \quad (3.7)$$

Then

$$\boxed{\begin{aligned} d_j \mathfrak{m}(X, Y) &= \mathfrak{m}(d_j X, P_{j-1} Y) + \mathfrak{m}(P_{j-1} X, d_j Y) \\ &+ \mathfrak{m}(d_j X, d_j Y) + \text{Cov}_j^{\mathfrak{m}}(X, Y) - \text{Cov}_{j-1}^{\mathfrak{m}}(X, Y). \end{aligned}} \quad (3.8)$$

Proof. Expand $P_j X = P_{j-1} X + d_j X$ and the analogous expression for Y inside the definition (3.7), then subtract the level $j-1$ identity. $\square$

For one Rademacher root, $X = Y = \xi$ and $P_0 \xi = 0$, one has

$$d_1(XY) = 0, \quad (d_1 X)(d_1 Y) = 1, \quad (3.9)$$

and the covariance increment in (3.8) equals -1 . Factorwise conditioning without this term is therefore false. For Gaussian roots,

$$\boxed{d_j \mathcal{Q} = G_{j-1} \otimes g_j + g_j \otimes G_{j-1} + g_j^{\otimes 2} - \Delta \Gamma_j}. \quad (3.10)$$

The rootwise Wick trace is again internal to the complete terminal variable.

The alternative moving-middle R-079 split differs from (3.5) by an exact transfer $\mathfrak{C}_j$. If $b_j = d_j \mathcal{J}_0$, $k_j = d_j \kappa$, $r_j = d_j \rho$, and $s_j = f_j - k_j = r_j - i_j$, then

$$\mathfrak{F}_j = \tilde{\mathfrak{F}}_j + \mathfrak{C}_j, \quad \mathfrak{I}_j = \tilde{\mathfrak{I}}_j - \mathfrak{C}_j, \quad (3.11)$$

where

$$\mathfrak{C}_j = \mathbb{E}\langle b_j + k_j, s_j \rangle + \frac{1}{2} \mathbb{E}\|s_j\|^2 + \frac{1}{2} \mathbb{E}(B_1 - B_j) : \Delta \Gamma_j. \quad (3.12)$$

The two orders agree on the complete block, but mixing their internal coefficient and future pieces loses or duplicates (3.12).

4 Predictable-baseline support collapse

After (2.10), consider

$$\mathfrak{P}_k = \mathcal{W}_{k-1}^{(J)}(A^{(k)}) - \mathcal{W}_{k-1}^{(J)}(A^{(k-1)}), \quad (4.1)$$

and put

$$Z_k = X_{k-1} + A^{(k-1)}, \quad G_k = DZ_k, \quad c_k = Da_k. \quad (4.2)$$

Theorem 4.1 (genuine large-gap residual is empty)

For the R-077 regular mutually orthogonal strict-past no-revisit whole-shell class, there is a fixed C_{supp} such that

$$\Delta_n G_k \neq 0 \quad \text{or} \quad \Delta_n (G_k^{\otimes 2} - \Gamma_{k-1}) \neq 0 \implies n \leq k + C_{\text{supp}}. \quad (4.3)$$

Every R-086 payload produced by the single shift a_k has largest-input label $r = k$. If

$$L_{\text{gap}} > C_{\text{supp}} + L_{\text{res}}, \quad (4.4)$$

then

$$\boxed{T_{Q,k}^> = T_{G,k}^> = 0.} \quad (4.5)$$

Proof. Whole-shell support gives (4.3), including the product support of the raw Wick factor. Resonance requires $|m - n| \leq L_{\text{res}}$, hence $m \leq k + C_{\text{supp}} + L_{\text{res}}$. This contradicts the coefficient-dominant requirement $m > k + L_{\text{gap}}$ under (4.4). $\square$

The coefficient $\Delta_m K_{t,k}$ need not be band limited: nonlinear composition may generate arbitrarily high m . The vanishing is caused by the cutoff of its resonant partner G_k or $G_k^{\otimes 2} - \Gamma_{k-1}$. If the historical symbol L_0 is used for both widths, a fixed $O(1)$ boundary collar may remain. Enlarging the payload-comparable branch by C_{supp} assigns that collar to the payable owner; it is not a genuine large-gap residual.

Consequently each rational predictable baseline reduces to

$$\boxed{\mathfrak{P}_{k,R} = \mathcal{U}_{R,k} + \mathcal{P}_{R,k}^{\text{pay}} + \frac{1}{2} c_k^T B_{1,k} c_k,} \quad (4.6)$$

up to that fixed payable collar. This is a support theorem, not yet a lower form estimate.

5 Conditional low-Hermite Wick compression

Condition on a sigma field $\mathcal{F}$ and write

$$G = \mu + g, \quad g \mid \mathcal{F} \sim N(0, \Sigma), \quad \Gamma = \Gamma_- + \Sigma. \quad (5.1)$$

For any square-integrable symmetric coefficient B , replace it inside the contraction by $\hat{B} = \mathbb{E}[B \mid \mathcal{F} \vee \sigma(g)]$ and let $\Pi_n \hat{B}$ denote the conditional Hermite projection in g .

Theorem 5.1 (only ranks zero, one, and two survive)

One has the exact conditional identity

$$\begin{aligned} \mathbb{E}_{\mathcal{F}}[B : (GG^T - \Gamma)] &= \langle \Pi_0 \hat{B}, \mu\mu^T - \Gamma_- \rangle \\ &\quad + \mathbb{E}_{\mathcal{F}}[\Pi_1 \hat{B} : (\mu g^T + g\mu^T)] \\ &\quad + \mathbb{E}_{\mathcal{F}}[\Pi_2 \hat{B} : (gg^T - \Sigma)]. \end{aligned} \quad (5.2)$$

Every coefficient Hermite rank $n \geq 3$ is orthogonal to the raw quadratic Wick factor.

Proof. Expand

$$GG^T - \Gamma = (\mu\mu^T - \Gamma_-) + (\mu g^T + g\mu^T) + (gg^T - \Sigma). \quad (5.3)$$

The three terms belong respectively to the zeroth, first, and second Gaussian chaoses in g . Conditional Hermite orthogonality proves (5.2). $\square$

Put

$$\beta_n^2 = \mathbb{E}_{\mathcal{F}} \|\Pi_n \hat{B}\|_{\text{HS}}^2, \quad \kappa_0 = \|\mu\mu^T - \Gamma_-\|_{\text{HS}}, \quad (5.4)$$

$$\kappa_1^2 = 2|\mu|^2 \text{tr} \Sigma + 2\mu^T \Sigma \mu, \quad \kappa_2^2 = (\text{tr} \Sigma)^2 + \text{tr}(\Sigma^2). \quad (5.5)$$

Block Cauchy–Schwarz gives

$$\left| \mathbb{E}_{\mathcal{F}}[B : (GG^T - \Gamma)] \right| \leq \sum_{n=0}^2 \beta_n \kappa_n \leq \left(\sum_{n=0}^2 \beta_n^2 \right)^{1/2} \left(\sum_{n=0}^2 \kappa_n^2 \right)^{1/2}. \quad (5.6)$$

With conditional Gaussian $W^{2,1}$ regularity, two integrations by parts give the equivalent Stein form

$$\mathbb{E}_{\mathcal{F}}[B : (GG^T - \Gamma)] = \overline{B} : (\mu\mu^T - \Gamma_-) + 2\mu \cdot \mathbb{E}_{\mathcal{F}} \text{div}_{\Sigma} B + \mathbb{E}_{\mathcal{F}} \text{div}_{\Sigma}^2 B. \quad (5.7)$$

Similarly,

$$q = \mathbb{E}_{\mathcal{F}}(BG) = \overline{B}\mu + \mathbb{E}_{\mathcal{F}} \text{div}_{\Sigma} B. \quad (5.8)$$

For $C = \Theta_R(B)^{1/2}$, whenever the same regularity is justified,

$$r = \mathbb{E}_{\mathcal{F}}(CG) = \overline{C}\mu + \mathbb{E}_{\mathcal{F}} \text{div}_{\Sigma} C. \quad (5.9)$$

The theorem also tensorizes without differentiating an adapted coefficient. Let independent Gaussian roots d_j build

$$G_j = G_{j-1} + d_j, \quad \Gamma_j = \Gamma_{j-1} + \Delta \Gamma_j. \quad (5.10)$$

Then

$$Q_J = Q_{\ell} + \sum_j \{2G_{j-1} \odot d_j + (d_j^{\otimes 2} - \Delta \Gamma_j)\}. \quad (5.11)$$

If $\Pi_n^{(j)}$ is the coordinate Hermite projection in d_j , conditioned on all other roots, then

$$\begin{aligned} \mathbb{E}[B : Q_J] &= \mathbb{E}[B : Q_{\ell}] + \sum_j \{ \mathbb{E}[\Pi_1^{(j)} B : 2G_{j-1} \odot d_j] \\ &\quad + \mathbb{E}[\Pi_2^{(j)} B : (d_j^{\otimes 2} - \Delta \Gamma_j)] \}. \end{aligned} \quad (5.12)$$

Componentwise,

$$\mathbb{E}(BG_J) = \mathbb{E}(BG_{\ell}) + \sum_j \mathbb{E}[(\Pi_1^{(j)} B) d_j]. \quad (5.13)$$

Deterministic spatial projections commute with these coordinate Hermite projections. The identities compress the probability chaoses but do not bound the surviving spatial fields.

6 Mean ownership and the full square

In the centered one-root case, the raw Wick term sees only $\Pi_2 B$, while q sees $\Pi_1 B$ and r sees $\Pi_1 C$. Hence the R-063 forest, whose owner is the raw Wick product, cannot cancel the two conditional means.

Let $\bar{A} = \bar{B} + 2R$ and define

$$\mathfrak{D}_R = \mathbb{E}_{\mathcal{F}}[B : \mathcal{Q}] - q^T \bar{A}^{-1} q - |r|^2. \quad (6.1)$$

The retained full-square block is exactly

$$\begin{aligned} \mathcal{S}_0(c) &= \frac{1}{2} \|c + \bar{A}^{-1} q\|_{\bar{A}}^2 + \frac{1}{2} \{ \mathbb{E}_{\mathcal{F}}[B : \mathcal{Q}] - q^T \bar{A}^{-1} q \} \\ &= \frac{1}{2} \|c + \bar{A}^{-1} q\|_{\bar{A}}^2 + \frac{1}{2} \mathfrak{D}_R + \frac{1}{2} |r|^2. \end{aligned} \quad (6.2)$$

Thus the transformed-mean debt r is restored by the retained terminal mean energy. The coefficient mean q remains coupled to the completed c square.

Fixture 6.1 (raw Wick zero, both means nonzero)

Let $g \sim N(0, 1)$ and

$$B(g) = 2 + \frac{1}{2} \tanh g, \quad R = 1. \quad (6.3)$$

This is smooth and strictly positive. Parity and Stein's identity give

$$\mathbb{E}[B(g)(g^2 - 1)] = 0, \quad q = \frac{1}{2} \mathbb{E} \operatorname{sech}^2 g > 0. \quad (6.4)$$

Since

$$C(g) = \left(\frac{2B(g)}{B(g) + 2} \right)^{1/2} \quad (6.5)$$

is strictly increasing, $r = \mathbb{E}[gC(g)] = \mathbb{E}C'(g) > 0$. The primary executable recomputes

$$q = 0.302852754801081\dots, \quad r = 0.0385455306276297\dots, \quad (6.6)$$

$$\mathfrak{D}_R = -0.0244157057040153\dots, \quad \inf_c \mathcal{S}_0(c) = -0.0114649738863249\dots \quad (6.7)$$

The raw owner is exactly zero while both mean sectors survive. Equation (6.2), not an additional forest correction, restores the r contribution.

7 Why low Hermite does not yet close the sum

For an adapted coefficient $C(\xi, A(\xi))$, Gaussian integration by parts gives

$$\mathbb{E}[\xi C] = \mathbb{E}[C_{\xi} + C_A A'], \quad (7.1)$$

and

$$\mathbb{E}[(\xi^2 - 1)C] = \mathbb{E}[C_{\xi\xi} + 2C_{\xi A} A' + C_{AA}(A')^2 + C_{AA} A'']. \quad (7.2)$$

The independent executable checks exact polynomial fixtures in which each selector term in (7.1)–(7.2) is nonzero. Existing strict-past one-shot theorems control Doob differences; they do not control these same-root Malliavin derivatives of later controls.

Proposition 7.1 (Hermite compression supplies no gain)

For $h_L(\xi) = -\tanh(L\xi)$ define

$$\mu_L = -\mathbb{E}[\xi h_L] = L\mathbb{E} \operatorname{sech}^2(L\xi), \quad \nu_L = \mathbb{E} h_L^2. \quad (7.3)$$

Then

$$\mu_L \longrightarrow \sqrt{2/\pi}, \quad \nu_L \longrightarrow 1, \quad (7.4)$$

and

$$\min_{\lambda} \{-\lambda\mu_L + \frac{1}{2}\lambda^2\nu_L\} = -\frac{\mu_L^2}{2\nu_L} \longrightarrow -\frac{1}{\pi}. \quad (7.5)$$

On the other hand,

$$\mathbb{E}|h'_L|^2 \sim \frac{4}{3\sqrt{2\pi}}L. \quad (7.6)$$

Proof. Dominated convergence gives (7.4); completing the scalar square gives (7.5). For (7.6), set $u = L\xi$ and use $\int_{\mathbb{R}} \operatorname{sech}^4 u \, du = 4/3$. $\square$

There is likewise no deterministic spatial gain. On the unit circle let

$$\phi_N(x) = \sqrt{2} \cos(Nx), \quad B_N(x, g) = \phi_N(x)H_2(g), \quad Q_N(x, g) = \phi_N(x)H_2(g). \quad (7.7)$$

For every integer N ,

$$\|\phi_N\|_2^2 = 1, \quad \mathbb{E} \int B_N Q_N = 2. \quad (7.8)$$

The second Hermite projection leaves the full spatial carrier untouched. Thus the exact useful reduction is a weighted block target such as

$$\begin{aligned} \left| \sum_{\alpha} \mathbb{E} \langle F_{\alpha}, Q_{\alpha} \rangle \right| &\leq \sum_{s=0}^2 \left(\sum_{\alpha} w_{\alpha} \|\Pi_s F_{\alpha}\|_2^2 \right)^{1/2} \\ &\quad \times \left(\sum_{\alpha} w_{\alpha}^{-1} \|\Pi_s Q_{\alpha}\|_2^2 \right)^{1/2}, \end{aligned} \quad (7.9)$$

provided a production-specific spatial weight can actually be proved.

8 The adapted-prefix frontier

The support collapse (4.5) leaves

$$Z_k = X_{k-1} + A^{(k-1)} \quad (8.1)$$

as a moving base correlated with the past Gaussian field. The decorated model in the remaining payable branch contains

$$D^3 \overline{B}_{k-1}^{(J)}(Z_k) \diamond Q_k. \quad (8.2)$$

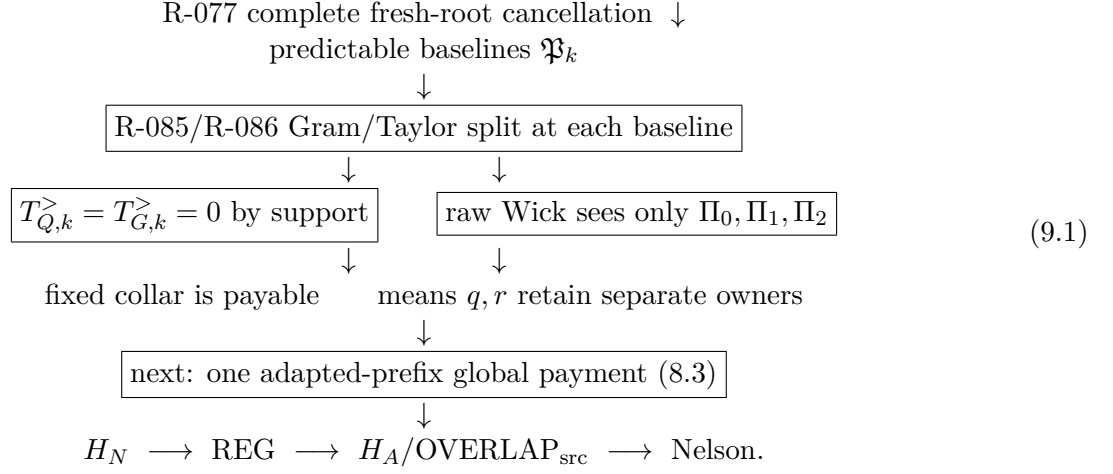
R-063 proves deterministic H^2 translation and reconstruction. It does not turn (8.2) into a control-independent model norm. Applying Young's inequality separately for every k would repeatedly consume the same prefix energy and terminal sextic budget.

The smallest honest successor is therefore

$$\boxed{\sum_k \mathbb{E} \left[\mathcal{U}_{R,k} + \mathcal{P}_{R,k}^{\text{pay}} + \frac{1}{2} c_k^T B_{1,k} c_k \right] \geq -\eta \mathbb{E} X - \zeta \mathbb{E} Y - C_{\eta, \zeta},} \quad (8.3)$$

assembled with every non-rational predictable-baseline owner and one global allocation. A sufficient new input would be a weighted conditional estimate for Π_0, P_{i_1}, P_{i_2} with a positive production root/shell gain. The alternative is a direct signed Schur/cancellation estimate that avoids Malliavin differentiation.

9 Updated proof and failure map



Route	Verdict	Reason / successor
Project rational row before R-077	failed	Cancellation is proved only for the complete Wick/trace/forest packet.
Factorwise Doob Leibniz	failed	The conditional covariance increment (3.8) is indispensable.
Predictable large-gap resonance	eliminated	Support and a widened fixed collar give (4.5).
Support collapse implies payment	failed	The moving adapted base (8.1) is outside the existing deterministic-translation theorem.
Low-Hermite compression	advanced	Only ranks zero through two survive, without differentiating the coefficient.
Stein/Doob closure	failed by present inputs	Selector derivatives survive and no spatial gain is created.
Global baseline sum	open / required	Prove (8.3) with all owners used once.

10 Devil's-advocate self-test

- Objection: Once $T_{Q,k}^> = T_{G,k}^> = 0$, the rational predictable baseline is proved. UPHELD as false.** Support removes only the genuine large-gap resonance. The remaining families live at the moving adapted base and their one-use global sum is not an existing theorem.
- Objection: The rational row may be projected before R-077 cancellation. UPHELD as false.** R-077 cancellation requires the complete Wick/trace/forest endpoint packet. Its individual rational or Hermite pieces need not be centered.
- Objection: Doob differences commute with Littlewood–Paley projections, so factorwise conditioning is legitimate. UPHELD as false.** Spatial projections commute, but bilinear evaluation produces the covariance defect (3.8). Fixture (3.9) has unit error if it is omitted.
- Objection: Only three Hermite ranks remain, so the coefficient is automatically small. UPHELD as false.** Orthogonality deletes high coordinate ranks but does not bound the surviving projections or change their spatial frequency; fixture (7.7)–(7.8) is frequency independent.
- Objection: Gaussian integration by parts converts the low Hermite projections to the accepted Doob square. UPHELD as false.** Equations (7.1)–(7.2) differentiate the selector. The tanh family retains an order-one negative bracket while its derivative energy diverges.

6. **Objection: The R-063 forest cancels q and r . UPHOLD as an ownership error.** The forest reconstructs raw Wick. The smooth fixture has raw Wick zero with both means nonzero; r is restored only by the retained terminal mean energy in (6.2).
7. **Objection: The two method fixtures disprove complete H_N . DISMISSED.** They omit complete signed companion terms and refute only support-implies-payment and unweighted Stein/Doob shortcuts. A direct signed global estimate remains possible.

11 Executable evidence

The primary executable uses non-unit Gaussian Hermite quadrature, an exact three-root tensor product, the smooth positive tanh mean fixture, bounded selector and spatial-carrier families, support enumeration, and a product commutator fixture. The non-importing independent executable uses exact Fraction product-Hermite coefficient ledgers, exact selector-chain and Rademacher identities, and a standard-library adaptive Simpson integrator. The integrated verifier runs both children afresh, hash-pins all authorities, checks the note, PDF, manifest, registry, explorations, claim and public surfaces, and fails closed on any promotion.

Run from the repository root:

```
python codes/foundations/
a13_classii_low_hermite_wick_predictable_baseline_reduction_verify.py
```

12 Result footer

Result ID: A13-CLASSII-LOW-HERMITE-WICK-PREDICTABLE-BASELINE-REDUCTION

Precise statement: Complete cancellation precedes Gram projection. On each predictable baseline the genuine large-gap T_Q and T_G sets are support-empty after a fixed payable collar. Raw Wick sees only coordinate Hermite ranks 0--2; q and r retain distinct rank-one/full-square ownership.

Scope: Finite cutoff, positive floor, R-077 regular strict-past no-revisit whole-shell class, widened fixed collar, and exact R-063/R-079/R-085/R-086/R-095 conventions.

Dependencies: R-063; R-077; R-079; R-085; R-086; R-095.

Evidence grade: ANALYTIC, EXACT, EXECUTED, REDUCTION, BOUNDARY.

Reproduction command: python codes/foundations/
a13_classii_low_hermite_wick_predictable_
baseline_reduction_verify.py

Expected output: Primary 69, independent 50, integrated manifest-pinned, and aggregate manifest-pinned assertions PASS; exit zero.

Falsification gate: Any identity, support collar, Hermite/mean ownership, fixture, authority, hash, PDF, count, public-link, or T4 scope assertion fails.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: Adapted-prefix payment, complete H_N , REG, H_A , OVERLAP_src, Nelson, removals, measure, Sector A, and T5--T7 remain open.

Next required action: Prove one global adapted-prefix payable form (8.3), using either weighted $\Pi_0/\Pi_1/\Pi_2$ spatial gain or a direct signed cancellation, with every owner used once.